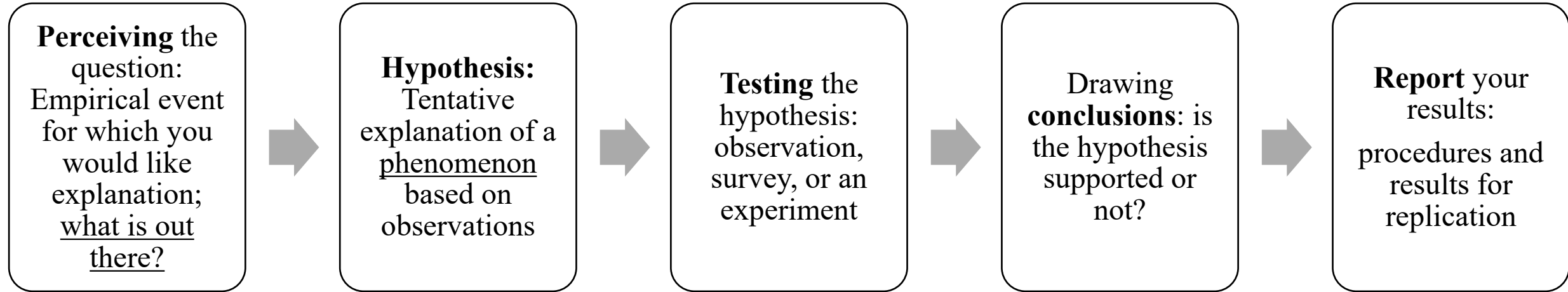




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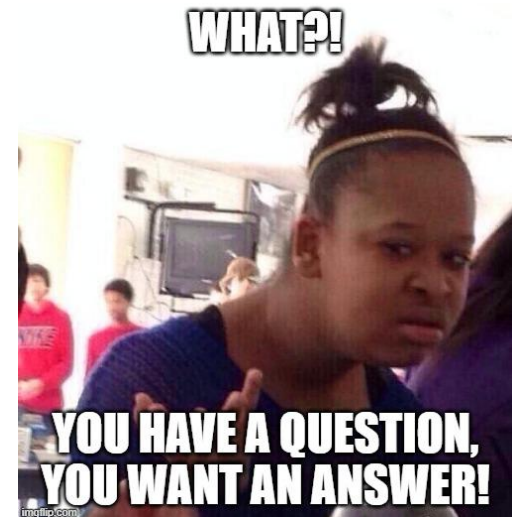
What is this Scientific Method?



Question: Does frequent use of Instagram impact the attention span of Gen Z college students?

Hypothesis: Increased daily use of Instagram is associated with shorter attention spans in Gen Z college students

(Ciccarelli & Meyer, 2006; Feldman, 2009)



Methods in Psychology



(Ciccarelli & Meyer, 2006)

Descriptive Method

- Systematic approach used to **observe**, **describe**, and **document** aspects of human behaviour, phenomena, or populations as they naturally occur
- Key Feature: Focus on observations
- Types of descriptive research
 - Naturalistic observation
 - Laboratory observation
 - Clinical case study
 - Survey
 - Cross-sectional and Longitudinal research

Descriptive Method: Naturalistic Observation

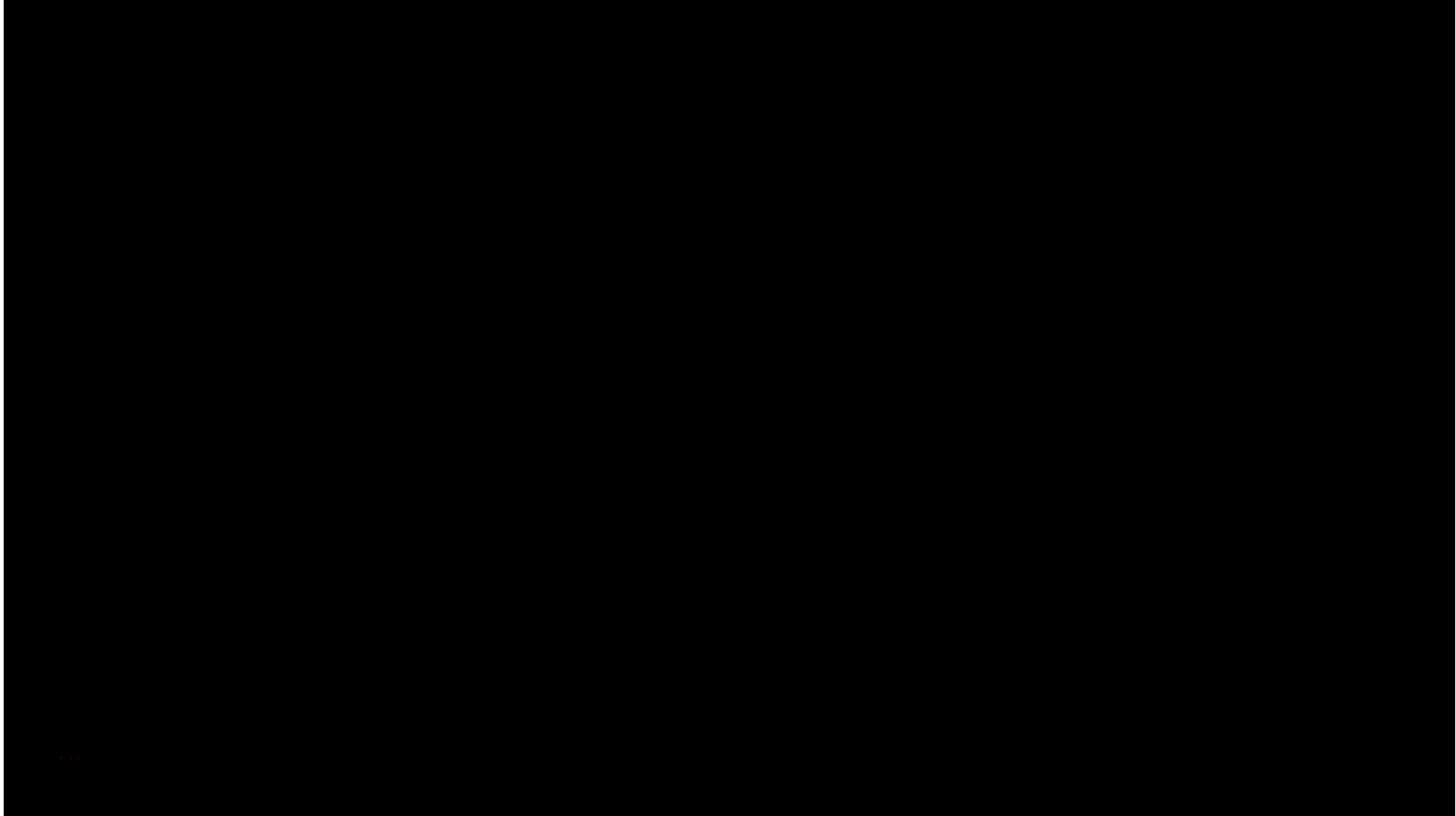
- Behaviour(s) of people and animal in their **natural environment**
- **Observer effect**: tendency to behave differently when ‘they’ know that they are being observed
- **Participant observation**: researcher becomes a participant in the group being observed
- Strength:
 - First-hand look
- Drawbacks:
 - Observer bias: tendency of researchers to see what they expect to see
 - Hawthorne effect: Increase in effective behaviour when being observed
 - Cannot be replicated

(Ciccarelli & Meyer, 2006)



“What is happening out there?”

Hawthorne effect



<https://www.youtube.com/watch?v=o4XX90lqT6E&t=22s>

Descriptive Method: Laboratory Observation

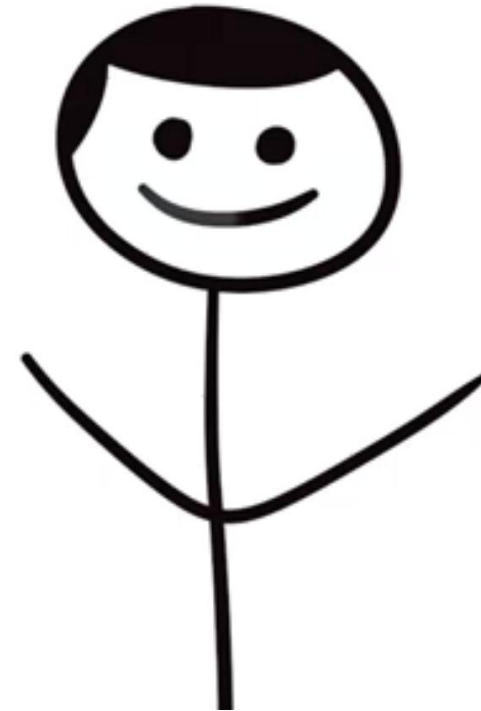
- Practicality issue is observing in natural setting
- Difficult in setting up screening equipment
- Trying to control various environmental factors



<https://www.youtube.com/watch?v=QTsewNrHUHU>

Descriptive Method: Clinical Case Study

- Individual studies in great details, try learning everything about one individual
- E.g., Phineas Gage
- Strength:
 - In-depth
- Drawback:
 - Cannot be replicated



(Ciccarelli & Meyer, 2006)

Descriptive Method: Survey

- Series of questions (mostly anonymous)
- Mode: in person, interview, or a questionnaire (online or offline)
- Covert Behaviours are assessed or screened
- **Representative sample**: randomly selected sample of participants from a larger population of participants
 - Example: Students of IIT Bombay
- Strength:
 - Can gather large data within a short period



Random: equal chance of being chosen



Descriptive Method: Survey (cont'd...)

<https://www.instagram.com/khapiter.guys>

15/01/2026

HS 110 | Introduction to Psychology



10

Descriptive Method: Survey (cont'd...)

Biases in survey

Courtesy bias:

Accuracy of response
(responding in socially
accurate manner)

What is socially desirable
Tests involves Lie scale or
repeated items

Wording bias:

The way question is
framed distorts the answers
than the belief of
individual

Examples: Leading
questions, Emotional
questions, Complex or
confusing words

Question order bias:

It occurs when sequence of
questions in a survey
influences how
respondents answer them.

For example: Priming
effect, fatigue effect, order
effect (leading to specific
way of answering)

Quick Recap!

- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
 - Descriptive
 - Correlational
 - Experimental
- Descriptive types
 - Naturalistic observation
 - Laboratory observation
 - Clinical case study
 - Survey